

Designing a Coherence-Gravity Test with Bose-Einstein Condensates

Identifiability, Measurement Error, Constraint Translation, and Go/No-Go Criteria

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Abstract

A Bose-Einstein condensate and a thermal cloud can contain the same isotope and nearly the same number of atoms while differing strongly in condensate fraction, spatial coherence, density profile, interaction energy, temperature, expansion, and preparation history. That combination makes a condensed-versus-thermal comparison scientifically attractive and causally difficult. This paper turns an earlier collaboration pitch into a quantitative experimental-design framework. The tested hypothesis is explicit: the Penrose-Onsager condensate density sources a neutral scalar through a dimension-four phenomenological operator. Inside a weak-field, linear, unscreened benchmark, that operator produces a differential acceleration proportional to the measured condensate-fraction contrast and an idealized interferometer phase proportional to the same contrast. The paper's new contribution is the identification architecture around that formula. It replaces a binary BEC / thermal label with a multi-level measured condensate fraction; derives the bias in a naive two-state contrast; incorporates measurement error in condensate fraction; defines a randomized blocked estimand and conditional sensitivity formula; specifies matched nuisance and negative-control families; and states a likelihood-based route from scalar models to existing constraints. The design is useful whether the hypothesis survives or fails. It yields a publishable go/no-go result when ordinary variables cannot be separated, when screening or prior constraints remove the accessible region, when the apparatus cannot reach the conditional effect, or when a preregistered measurement supports the null. It reports no measured anomaly and no constraint-cleared nonzero parameter point.

Scope and evidence boundary

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Reader Orientation: What Must Be Separated Before “Coherence” Can Be Tested

Imagine cooling two rubidium-87 clouds in the same apparatus. One cloud contains a large condensed component; the other is mostly thermal. A naive question asks whether gravity treats the two clouds differently. The real problem is harder: cooling through the transition changes several physical properties at once. If the measured phase changes, the cause could be condensate fraction, but it could also be temperature, density, mean-field interaction, cloud size, release velocity, magnetic response, optical sampling, or detection bias.

Established physics already explains why many of those ordinary changes matter. Interactions in a BEC produce nonlinear phase evolution and phase diffusion [8]. BEC interferometers can measure inertial and

differential forces, including in microgravity [4,9]. Atom interferometers have also constrained screened scalar-force models using controlled source masses [10-12]. None of those results establishes a new gravitational charge carried by condensate density.

The proposed operation can therefore be understood before it is named:

1. prepare many runs spanning several condensate fractions rather than only two labels;
2. reconstruct the condensate fraction and mode independently for every run;
3. measure ordinary state variables that change with the preparation;
4. predict how a declared condensate-dependent source would alter acceleration or phase;
5. randomize, blind, and fit the condensate-dependent term jointly with ordinary alternatives; and
6. require the result to follow the predicted sign, source geometry, range, and state dependence.

This is a **coherence-gravity test** only in its historical and motivating sense. The actual measured input is the Penrose-Onsager condensate fraction and density. The name summarizes a specific testable chain; it does not substitute the word *coherence* for an observable.

1. Research Question and Scope

The bounded question is:

After controlling and measuring ordinary changes caused by preparation, is there a reproducible differential acceleration or interferometer-phase component that follows condensate fraction in the way predicted by one declared condensate-source operator?

The same-isotope design preserves the best feature of the original proposal. It removes composition as the intended treatment contrast and focuses the experiment on preparation state. It does not remove all ordinary differences between the preparations. Those remaining differences are the central identification problem of this paper.

The paper has four positive outputs:

- a complete conditional map from condensate reconstruction to an instrument-native outcome;
- an estimand that can be identified only under stated experimental conditions;
- a constraint and sensitivity translation that cannot be confused with a measured bound; and
- explicit continuation and retirement criteria.

Clock, SuperTimePosition, space-platform, and gravitational-phase-lock experiments remain separate future protocol families. They are preserved from the original program but are not pooled into this paper's primary endpoint.

2. Condensate Reconstruction and the Declared Source Hypothesis

The condensate is defined operationally through the largest eigenpair of the one-body density matrix [5]:

$$\begin{aligned}
 \int \rho_1(\mathbf{x}, \mathbf{x}') \phi_0(\mathbf{x}') d^3x' &= N_0 \phi_0(\mathbf{x}), \\
 \int |\phi_0(\mathbf{x})|^2 d^3x &= 1, \\
 n_0(\mathbf{x}) &= N_0 |\phi_0(\mathbf{x})|^2, \\
 f_0 &= \frac{N_0}{N}.
 \end{aligned} \tag{1}$$

Here N_0 is the condensate occupation, ϕ_0 is the normalized condensate mode, n_0 is its number-density field, and f_0 is condensate fraction. The scalar amplitude σ is not identified with f_0 , visibility, purity, or coherence length.

The inherited Level 0 model [16] adds the phenomenological operator

$$\mathcal{L}_Q = -\frac{g_Q m_a}{M_{\text{Pl}}} \sigma n_0, \quad (2)$$

where m_a is atomic mass and g_Q is a dimensionless coupling. In four-dimensional natural units, $[\sigma] = 1$ and $[n_0] = 3$, so the operator has mass dimension four. This dimensional consistency does not supply a microscopic derivation. A complete source theory would still need the condensate current, stress tensor, exchange with the noncondensed fraction, trapping terms, and backreaction.

The bounded static linear equation is

$$(\nabla^2 - m_\sigma^2)\sigma = \frac{\beta_\sigma \rho + g_Q m_a n_0}{M_{\text{Pl}}}. \quad (3)$$

For a preparation with condensate fraction f_0 , the effective test-cloud scalar coupling in this benchmark is

$$\beta_{\text{cloud}}(f_0) = \beta_\sigma + g_Q f_0. \quad (4)$$

Using $M_{\text{Pl}}^{-2} = 8\pi G$, a source with effective scalar coupling β_S , and a point-source Yukawa response $Y(x) = (1+x)e^{-x}$, the matched-preparation differential response is

$$\eta_Q = \frac{a_1 - a_2}{a_N} = 2\beta_S g_Q (f_{0,1} - f_{0,2}) Y(x), \quad x = m_\sigma r. \quad (5)$$

For a linear, unscreened, uniform spherical source of radius R observed at height h , define $z = m_\sigma R$ and $y = m_\sigma h$. The exact exterior response is

$$Y_{\text{sphere}}(z, y) = \frac{3(1+z+y)}{2z^3} \left[(z-1)e^{-y} + (z+1)e^{-2z-y} \right]. \quad (6)$$

An idealized light-pulse interferometer maps the response to phase as

$$\Delta\phi_Q = k_{\text{eff}} g \eta_Q T^2. \quad (7)$$

Equations (1)-(7) form a software-checkable conditional chain. They do not establish that Equation (2) exists in nature, that a source is unscreened, or that any nonzero coupling survives existing data.

3. Why a Binary BEC / Thermal Contrast Does Not Identify the Coupling

Let A denote the preparation label, f_0 the true condensate fraction, $\mathbf{C}$ a vector of ordinary state and apparatus variables, and Y the measured phase or acceleration outcome. Suppose the local response can be written as

$$Y = \alpha + \gamma_Q f_0 + \bar{\delta}^T \mathbf{C} + \varepsilon. \quad (8)$$

The difference between average BEC-labeled and thermal-labeled runs is then

$$\Delta_A = E[Y \mid A = 1] - E[Y \mid A = 0] = \gamma_Q \Delta_A f_0 + \bar{\delta}^T \Delta_A \mathbf{C} + \Delta_A E[\varepsilon \mid A]. \quad (9)$$

The first term is the desired condensate-dependent component. The second collects temperature, density, interaction energy, expansion, spatial sampling, magnetic response, and other ordinary changes. The third captures unmodeled preparation and selection effects. A binary contrast alone cannot determine which term produced the difference.

Randomizing the order of two preparation labels protects against drift but does not independently randomize condensate fraction, temperature, density, and interaction energy. Identification therefore requires variation that changes these quantities in partially independent directions. The minimum design uses at least three prespecified condensate-fraction levels and dedicated control families:

Control family	Intended variation	Intended near-match	Question answered
Condensate ladder	f_0 across a broad range	isotope, apparatus, block, geometry	Does outcome track measured f_0 ?
Thermal nuisance control	temperature and thermal fraction	f_0 as closely as feasible	Can thermal changes imitate the slope?
Density / interaction control	density and mean-field energy	f_0 and temperature	Can interactions imitate the slope?
Expansion / optical control	cloud size and release history	reconstructed f_0	Can beam sampling or launch history imitate it?
Source-geometry control	source position or geometry	atomic preparation	Does a residual follow the predicted field geometry?
Null channel	instrument channel insensitive to proposed coupling	drift sensitivity	Does the effect occur where it should not?

If the apparatus cannot produce sufficient independent variation, the design is not identifiable. That is a substantive result and a reason to retire or redesign the route.

4. Condensate-Fraction Measurement Error

The theory uses the true f_0 , but the experiment reconstructs an estimate. Write

$$\tilde{f}_{0,br} = f_{0,br} + u_{br}, \quad E[u_{br} \mid f_{0,br}, \mathbf{C}_{br}] = 0, \quad (10)$$

where b indexes randomized apparatus blocks and r indexes runs. In the simplest independent classical-error case, regressing Y on $\tilde{f}_0$ attenuates the slope by the reliability ratio

$$\lambda_f = \frac{\text{Var}(f_0)}{\text{Var}(f_0) + \text{Var}(u)}, \quad (11)$$

$$E[\hat{\gamma}_{\text{naive}}] \approx \lambda_f \gamma_Q.$$

The correction is not to divide blindly by an estimated λ_f . The analysis should fit a latent- f_0 or errors-in-variables model using calibration data, repeated reconstructions, or a validated image-to-condensate measurement likelihood:

$$p(Y, \tilde{f}_0 | \theta) = \int p(Y | f_0, \mathbf{C}, \theta) p(\tilde{f}_0 | f_0, \mathbf{C}, \theta) p(f_0 | \mathbf{C}, \theta) df_0. \quad (12)$$

This matters twice. Measurement error can hide a real state dependence by attenuation, and preparation-dependent reconstruction error can create a false state dependence. The image processing version, thresholds, exclusions, and uncertainty model must therefore be frozen before the outcome labels are opened.

5. Blocked Estimand and Conditional Sensitivity

A preregistered linear working model can take the form

$$Y_{br} = \alpha_b + \gamma f_{0,br} + \vec{\delta}^\top \mathbf{C}_{br} + \vec{\zeta}^\top (f_{0,br} \mathbf{C}_{br}) + \epsilon_{br}. \quad (13)$$

α_b absorbs block-specific common shifts. The nuisance vector should include, at minimum, atom number, temperature, density profile, interaction-energy proxy, center of mass, cloud shape, release velocity, magnetic state, field gradients, laser intensity, wavefront sampling, vibration, gravity gradient, Coriolis terms, detection quality, and run order. Nonlinear or hierarchical extensions may be prespecified, but a flexible model chosen after outcome inspection cannot define the confirmatory result.

The idealized physical slope predicted by Equations (5)-(7) is

$$\gamma_Q(\theta) = 2k_{\text{eff}} g T^2 \beta_{sgQ} Y_{\text{source}}(m_\sigma, \mathcal{G}), \quad (14)$$

where $\mathcal{G}$ denotes source geometry and Y_{source} is the justified point, uniform-sphere, layered-source, or numerically screened response. In the homoscedastic single-regressor limit, the slope uncertainty scales as

$$\text{SE}(\hat{\gamma}) \approx \frac{\sigma_Y}{\sqrt{\sum_{i=1}^n (f_{0,i} - \bar{f}_0)^2}}. \quad (15)$$

Equation (15) makes a design principle explicit: precision depends not only on run count but on the spread of independently measured condensate fractions. Repeating nearly identical f_0 values cannot identify a state-response slope efficiently.

For a prespecified two-sided threshold $z_\star$, a conditional minimum detectable coupling product is

$$|\beta_{sgQ}|_{\min} = \frac{z_\star \text{SE}(\hat{\gamma})}{2k_{\text{eff}} g T^2 |Y_{\text{source}}|}. \quad (16)$$

This is a design equation, not a reported limit. It becomes an experimental sensitivity only after σ_Y , covariance, f_0 reconstruction error, transfer functions, missingness, and screening are calibrated from the selected apparatus.

6. Translating Existing Constraints

Equivalence-principle and fifth-force experiments establish a demanding comparison environment [1-3,6,7,10-12]. They do not directly report a bound on g_Q , because they generally vary composition, internal state, distance, or source geometry rather than Penrose-Onsager condensate fraction. A valid translation begins with one shared parameter vector

$$\theta = \{m_\sigma, g_Q, \beta_S, \beta_\sigma, \text{screening parameters, source parameters, apparatus nuisances}\}. \quad (17)$$

Each experiment j needs its own forward map $M_j(\theta, \nu_j)$, nuisance vector ν_j , and reported-data likelihood. The combined constraint is

$$\mathcal{L}_{\text{joint}}(\theta) = \prod_j \int \mathcal{L}_j(D_j \mid M_j(\theta, \nu_j)) \pi_j(\nu_j) d\nu_j. \quad (18)$$

Only after those maps are derived may one define an allowed set, for example

$$\mathcal{A}_\alpha = \left\{ \theta : -2 \log \frac{\mathcal{L}_{\text{prof}}(\theta)}{\mathcal{L}_{\text{prof}}(\hat{\theta})} \leq q_\alpha \right\}. \quad (19)$$

The exact threshold and construction must match the chosen frequentist or Bayesian procedure. Importing a published acceleration sensitivity as though it directly constrained g_Q would skip the source charge, screening, geometry, and observable maps.

Atom-interferometric searches for chameleon and symmetron forces demonstrate why the full translation matters. Hamilton et al., Jaffe et al., and Sabulsky et al. modeled the source, vacuum geometry, screening assumptions, atomic probe, and measured acceleration before drawing model-specific exclusions [10-12]. The condensate-source hypothesis needs the same level of specificity. If the resulting $\mathcal{A}_\alpha$ has no overlap with the apparatus-reachable region, the BEC route should be closed before experimental outreach.

7. Apparatus and Run Contract

The protocol retains one isotope and one apparatus while deliberately varying measured condensate fraction. Each block randomizes preparation order and includes calibration or reference runs. Blinding protects the mapping from preparation labels to the outcome-analysis team until data quality, exclusions, and preprocessing are frozen.

The primary outcome should be one instrument-native differential phase or acceleration statistic. Secondary channels may diagnose ordinary mechanisms but must not replace the primary endpoint after inspection.

Domain	Required measurement or control	Failure if absent
Atomic state	isotope, hyperfine state, atom number, condensate fraction	intended treatment undefined
Thermodynamics	temperature, thermal fraction, density, interaction proxy	BEC label confounded
Kinematics	center of mass, launch, release, velocity, expansion	acceleration / phase mimic unresolved
Optical system	intensity, detuning, wavefront, AC-Stark contribution	state-dependent optical bias unresolved

Domain	Required measurement or control	Failure if absent
Magnetic system	field and gradients, Zeeman sensitivity	magnetic force mimic unresolved
Inertial environment	vibration, rotation, Coriolis, gravity gradient	common and differential acceleration unresolved
Detection	imaging transfer, fit quality, saturation, missingness	f_0 and phase bias unresolved
Source field	mass map, position, modulation, range response	geometry prediction untestable

The all-runs ledger must retain failed preparations, invalidated runs, exclusions, censoring, and reason codes. Missingness must be modeled or bounded when it depends on preparation or outcome quality.

8. Positive, Null, and Adverse Outcomes

A result is not classified by whether it is exciting. It is classified by which preregistered relations survive.

8.1 Positive-pattern requirement

A candidate update to the model must satisfy all of the following:

- a nonzero γ with the prespecified sign convention;
- stability across blocks and allowed nuisance models;
- no corresponding signal in negative-control channels;
- scaling with independently reconstructed f_0 , not merely the BEC label;
- the predicted source-position, geometry, or range dependence;
- survival of label permutation and analysis-pipeline controls; and
- replication in an independently configured dataset or apparatus.

A surviving residual would update the declared operator relative to the tested alternatives. It would not, by itself, validate Super Information Theory as a whole.

8.2 Null and exclusion requirement

For a validated forward map and likelihood, the experiment can exclude the subset

$$\mathcal{E}_\alpha = \{\theta \in \mathcal{A}_{\text{prior}} : \text{the prespecified test rejects } M(\theta) \text{ at level } \alpha\}. \quad (20)$$

Without the forward map, a null is still useful as an apparatus and design result, but it is not a sharp theory exclusion.

8.3 Adverse design results

The route should be retired or redesigned if:

1. no covariantly or microscopically defensible source approximation can be stated;
2. screening and existing constraints leave no apparatus-reachable region;
3. condensate fraction cannot be varied separately enough from ordinary variables;
4. condensate-fraction reconstruction error dominates the useful range;
5. calibrated nuisance covariance makes Equation (16) unreachable;
6. control channels reproduce the candidate slope;

7. the effect fails source-geometry, range, sign, or replication tests; or
8. the preregistered data support the null over the reachable region.

These are scientific outputs, not administrative failures.

9. Go/No-Go Sequence

The original collaboration pitch is replaced by a staged decision architecture:

Stage	Required evidence	Continue when	Stop when
1. Source definition	Equations (1)-(3), dimensions, conservation boundary	bounded source model is coherent	no defensible source approximation exists
2. Constraint translation	Equations (17)-(19), source geometry, screening	nonempty accessible region survives	region is empty or translation fails
3. Apparatus model	pulse sequence, transfer functions, calibrated covariance	conditional reach overlaps allowed region	apparatus floor dominates
4. Identifiability	condensate ladder and nuisance controls	f_0 has independent support	state variables remain inseparable
5. Analysis lock	estimand, missingness, multiplicity, decision thresholds	protocol is frozen before labels open	outcome-dependent flexibility remains
6. Execution	blinded randomized data and all-runs ledger	quality gates pass	prespecified failure criterion triggers
7. Interpretation	controls, geometry, sign, range, replication	complete pattern survives	null or ordinary mechanism explains data

Specialist criticism can improve any future version, but the local final-preprint result is this complete design and failure architecture. Conducting the experiment is not a prerequisite for publishing the architecture; it is the future use for which the architecture is intended.

10. Reproducibility Surface

The accompanying machine-readable protocol performs deterministic checks of:

- upstream Level 0 manuscript and result identities;
- the dimensional and null-limit contracts;
- the finite-source response;
- the binary-confounding decomposition;
- measurement-error attenuation;
- slope-precision scaling;
- conditional coupling sensitivity;
- open constraint and apparatus boundaries;
- manuscript equation and reference sequences; and
- the exact release package.

Numerical regression fixtures test algebra and software. They are not measurements, forecasts, or allowed parameter points. The public package includes the manuscript, model inputs, verifier, tests, source and correction ledgers, metadata, rendering tool, review receipts, and exact SHA-256 manifest.

11. Relationship to SIT and the Earlier Pitch

The earlier paper contributed the bounded experimental intuition: compare a coherent and a noncoherent preparation of the same species rather than ask an experiment to validate an entire unification program. That intuition is preserved. The successor replaces three insufficient elements:

- the undefined $F(\dots)$ is replaced by Equations (1)-(7);
- the binary state label is replaced by a measured condensate-fraction estimand; and
- a generic sensitivity request is replaced by Equations (16)-(20) and the go/no-go sequence.

The source operator remains an explicit Level 0 hypothesis rather than a derivation from SIT79. The experiment, if performed, tests that operator and its declared response. It does not test every claim associated with SIT, SDT, QGTCD, or SuperTimePosition simultaneously.

12. Conclusion

A same-isotope BEC / thermal experiment becomes informative only after the preparation label is decomposed into measurable physical variables. The proposed condensate-source operator supplies a clear conditional prediction: within its stated weak-field benchmark, differential acceleration and phase are linear in condensate-fraction contrast and follow a specified source response. The experimental-design problem is then to distinguish that slope from the ordinary physical changes that accompany condensation.

This paper provides the needed bridge from idea to test. It formalizes the confounding decomposition, measurement-error model, blocked estimand, conditional sensitivity, constraint translation, control families, likelihood boundary, and retirement rules. The result is useful before any laboratory commits time: it can show that the route is viable, that it needs redesign, or that it should be closed. A future experiment can then produce a positive-pattern update, a model-specific exclusion, or a bounded null without changing the question after seeing the data.

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Release Boundary

This Version 2 final preprint is locally complete. No Zenodo version, OSF change, GitHub push, journal submission, collaborator communication, participant recruitment, or experimental action has yet been performed. The package enters the governed portfolio release sequence. A later experimental protocol requires apparatus-specific calibration, a completed constraint translation, a frozen analysis plan, and the relevant institutional and collaborator approvals.